

ECON XXX

Problem Set 1

Your Name

Due: September 26, 2026

Question 1

Write the question here, or paste only the part you need.

Solution

Suppose

$$y_i = \alpha + \beta x_i + \varepsilon_i, \quad i = 1, \dots, n. \quad (1)$$

A useful aligned derivation is

$$\mathbb{E}[y_i \mid x_i] = \mathbb{E}[\alpha + \beta x_i + \varepsilon_i \mid x_i] \quad (2)$$

$$= \alpha + \beta x_i + \mathbb{E}[\varepsilon_i \mid x_i]. \quad (3)$$

Therefore, if $\mathbb{E}[\varepsilon_i \mid x_i] = 0$,

$$\mathbb{E}[y_i \mid x_i] = \alpha + \beta x_i.$$

Question 2

Solution

For multiple parts, use an enumerated list:

- (a) First part.
- (b) Second part.
- (c) Third part.

Question 3

Solution

A small table can be written as

Variable	Mean	Std. Dev.
X	1.25	0.42
Y	3.18	1.07